Assignment: Designing a Video Generation Network—Your Solution versus an AI Solution

- Suppose you are tasked with developing a deep learning system that generates synthetic videos depicting Donald Trump for clearly labelled educational or research demonstrations. Develop your own proposal, use a generative AI tool to obtain an alternative proposal, and critically compare the two. The aim is to assess your technical understanding and your ability to evaluate AI-generated suggestions.

The solution will be in three parts:

Part 1 Before consulting a generative AI tool, describe your proposed approach, covering:

- **Data:** What data would you need? How would you collect, preprocess, and organise the training and evaluation datasets?
- **Network architecture:** What are the main components, inputs, and outputs? How would your model maintain identity and temporal consistency across frames?
- **Resources:** What hardware, software, pretrained models, and computational budget would you require?
- **Design rationale:** Explain your choices, assumptions, expected limitations, and trade-offs.

Part 2: AI-assisted proposal and critical comparison

- Use a generative AI tool to propose a solution to the same task and compare the AI-generated proposal with your independent proposal.
- Explain the main similarities and differences in data, architecture, resources, and evaluation.
- Identify at least one weaknesses, unsupported claims, or missing details in the AI proposal. Explain how you checked them.
- Identify at least one useful AI suggestion and explain whether you would adopt, modify, or reject it.

Part 3: Conclusion

- How do you think AI can help you develop and improve your proposed solution?

Please submit your work in PDF format, encouraged to be within 5 pages.

Online submission system close after 10 Nov 2026